DRILL 04

GROUND LADDERS

NFPA 1001 Level 1 (JPR 4.3.6, 4.3.12) / WorkSafeNB Reg 91-191

GROUND LADDERS – CARRIES, RAISES & PLACEMENT

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Ladder Rack | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | 2-Firefighter Teams

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to remove a 24-ft or 28-ft extension ladder from the apparatus, carry it safely using 1-person and 2-person methods, raise and extend it to a target window or roofline, lock the dogs, tie off the halyard, and set the proper 75-degree climbing angle. Ground ladders are the primary secondary egress for interior crews operating in two-story rural dwellings.

📦 TOOLS & APPARATUS SETUP

- Apparatus ground ladder complement: 24-ft or 28-ft two-section extension ladder, 14-ft roof ladder.
- Station exterior wall, drill tower, or suitable clear wall structure without overhead wires.
- Ladder staging area clear of potholes, gravel, or ice.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Overhead Electrical Wires:** Always look up **360 degrees** before removing or raising any ground ladder. Maintain minimum 3-metre (10-foot) clearance from power lines.

⚠️ **Pinch Points & Dog Locks:** Keep fingers wrapped around ladder beams—never grasp rungs while raising or lowering the fly section.

⚠️ **Footing & Heel Placement:** The "heeler" must place their steel-toed boots against the ladder spurs and pull back on the beams while the extension is raised.

🛑 **THE STOP RULE:** Shout **"STOP!"** immediately if the ladder begins to twist, if footing slips, or if anyone touches an overhead hazard.

PHASE 1: REMOVAL, INSPECTION & CARRIES (15 MINUTES) — SLOW DRY RUN

1. **Apparatus Removal:** Release ladder retaining bracket. 2 firefighters lift ladder and slide it clear of apparatus bed.
2. **Two-Firefighter Low-Shoulder Carry:**
   - Firefighter at the butt (heel) faces direction of travel. Firefighter at the tip faces direction of travel.
   - Both place upper arm between rungs, resting the beam on the shoulder.
   - Butt firefighter calls: *"Ready to lift... Lift! Ready to walk... Forward!"*
3. **One-Firefighter Low-Shoulder Carry (14-ft roof or 24-ft flat carry):**
   - Center on ladder balance point, reach through rungs, lift onto shoulder, stabilize tip with leading hand.

DRILL 04: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ground Ladders • Practical Application

PHASE 2: THE TWO-FIREFIGHTER EXTENSION LADDER RAISE (20 MINUTES) — BUDDY COACHING

1. **Positioning:**
   - Butt firefighter places ladder heel on ground at proper distance from wall (~6 feet for a 24' ladder raised to second story).
   - Butt firefighter stands on the bottom rung or places both boots against the spurs, grasping beams firmly at chest level.
2. **Beam Raise (or Flat Raise):**
   - Tip firefighter lifts ladder above head and walks hand-over-hand along the rungs/beams toward the butt, pushing ladder to vertical.
3. **Extension & Dog Lock:**
   - Butt firefighter stabilizes vertical ladder, eyes looking up.
   - Tip firefighter steps to halyard side, grasps rope hand-over-hand, and pulls smoothly to extend fly section.
   - Ensure fly is oriented according to manufacturer specification (usually **fly out** for modern aluminum extension ladders).
   - Once tip reaches 3–5 rungs above roofline (or sill level for rescue), lower slightly until **ladder dogs (pawls) click and lock onto rungs**.
4. **Halyard Tie-Off:**
   - Wrap halyard around the rung below the dogs and tie off with a **clove hitch** and half hitch around the beam.
5. **Setting Angle (1:4 Ratio / 75 Degrees):**
   - Butt firefighter stands straight with toes touching ladder spurs, reaches arms straight forward. Palms should comfortably rest on the ladder rung at shoulder height.

PHASE 3: LADDER PLACEMENT SCENARIOS (15 MINUTES) — TIMED / STRESS

Rotate crews through 3 tactical placement positions:

1. **Window Rescue:** Tip placed directly at or just below the windowsill for rapid victim removal.
2. **Ventilation / Entry:** Tip placed upwind side of window, level with upper third of window frame.
3. **Roof Access:** Tip extended minimum **3 to 5 rungs above roof eaves** to provide visible handhold for roof crews in dark conditions.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Looking down while extending:* The heeler must keep their eyes on the tip to watch for wires and ensure dogs lock properly.
- ❌ *Fingers on rungs during lowering:* If halyard slips, fly drops instantly—shearing off fingers! Keep hands strictly on the beams.
- ❌ *Shallow climbing angle (<70°):* Causes ladder base to kick out under load. Always verify the arm-reach angle check.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did anyone forget the halyard knot?", "Was the climbing angle verified before stepping onto the bottom rung?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Fly section lowered smoothly; dogs inspected for spring tension.
- ☐ Halyard neatly coiled and tied to bottom rungs.
- ☐ Ladder beams wiped clean of grit and road salt.
- ☐ Ladder stowed into apparatus rack and locked with safety pins.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS